FLIGHT CONTROL SYSTEM SAFETY ASSESSMENT

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Table of contents

1 Introduction

This document presents the safety assessment of the flight control system for the fictitious modification used in this prototype.

1.1 Scope

The assessment covers the primary and secondary flight controls.

1.2 Applicable requirements

The assessment supports compliance with CS 25.671 and CS 25.1309(b) at Amdt 27, using AMC 25.1309 guidance.

2 System description

The system comprises fictitious computers, actuators and sensors.

2.1 Architecture overview

Three dissimilar computing lanes are assumed for demonstration purposes.

3 Functional hazard assessment

The functional hazard assessment identifies the failure conditions and their classification.

3.1 Failure condition list

Fictitious failure conditions FC-01 to FC-18 are considered.

3.2 Classification rationale

Classification follows the severity definitions of AMC 25.1309.

3.3 Functional hazard assessment results

All identified failure conditions are assessed as acceptable in this fictitious dataset. Results are summarised in table 3-2.

4 Safety analyses

This chapter consolidates the quantitative and qualitative analyses.

4.1 Fault tree analysis

Fictitious fault trees are built for each catastrophic failure condition.

4.2 Failure modes and effects analysis

The FMEA covers the actuator and sensor fictitious failure modes.

4.3 Common cause analysis

The common cause analysis addresses zonal, particular risks and common mode aspects.

4.3.1 Zonal safety analysis

Fictitious zonal analysis of the avionics bay and wing zones.

4.3.2 Particular risks analysis

Particular risks considered: fictitious bird strike, tyre burst and hydraulic fluid loss scenarios.

4.3.3 Common mode analysis

Common mode analysis of the three fictitious computing lanes.

4.4 Quantitative assessment

The fictitious computed probabilities meet the objectives of CS 25.1309.

5 Conclusion

On the basis of this fictitious assessment, the flight control system is considered compliant with CS 25.671 and CS 25.1309(b).

5.1 Open points

No open point in this fictitious dataset.